MODERN PHYSICS – MIDTERM 2. RETAKE

Please fill your **Neptun code** and **Name** on **all papers you submit**. Preferably use these papers for all your calculations as well. If using any extra sheets of paper you submit, write your Neptun code and name on both sides of those too.
Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. The score is written for each part of the problem as a sum. Bonus points are marked with a *, they are added to your score, but the total can't exceed 50 points. Some parts of the problems don't rely on the result in earlier parts, make sure to read every part of the problems.

Physical constants

$c = 3.00 \times 10^8\,\mathrm{m/s}$ $\qquad$ $h = 6.626\,070\,15 \times 10^{-34}\,\mathrm{J\,s} = 4.135\,667\,7 \times 10^{-15}\,\mathrm{eV\,s}$

$hc = 1.9864 \times 10^{-25}\,\mathrm{J\,m} = 1239.8\,\mathrm{eV\,nm}$ $\qquad$ $m_e c^2 = 511\,\mathrm{keV}$

$\lambda = \dfrac{h}{p} = \dfrac{2\pi}{k}$ $\qquad$ $\mu_B = e\hbar/2m_\mathrm{e} = 5.788\,38 \times 10^{-5}\,\mathrm{eV/T}$

1. Energy spectrum of atomic Oxygen (5 + 4 + 5 + 3 + 6* = 17 + 6* points)

Oxygen (O) has the ground state electron configuration $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^4$.

(a) Draw the configuration (with arrows on levels), and find the total S, L, J quantum numbers of the ground state of the O atom. Label the state using the spectroscopic notation.
Hint: Hund's rules are: (i) maximize S (ii) maximize L (iii) for less than half-filled shell, minimize J, for more than half-filled, maximize J.

(b) If a 2p electron is excited to the 3s orbital, we get an excited state of O with the electron configuration $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^3 3\mathrm{s}^1$. We consider the state with three spin up ($m_s = +1/2$) electrons on the 2p level, and one spin down ($m_s = -1/2$) electron on the 3s level. Find the total S, L, J quantum numbers in this state, and label the state in the spectroscopic notation.

(c) Observing the emission spectrum of O (without magnetic field) we find that the line corresponding to the transition $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^3 3\mathrm{s}^1 \rightarrow [\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^4$ has a fine structure: it is split into three closely spaced lines with wavelengths $\lambda_0 = 130.60\,\mathrm{nm}$, $\lambda_1 = 130.49\,\mathrm{nm}$, and $\lambda_2 = 130.22\,\mathrm{nm}$. This is because the $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^4$ configuration is spin-orbit split into three levels corresponding to different values of $J = 0, 1, 2$. (Only one of these is the ground state satisfying Hund's third rule, the others are excited states with slightly higher energy.) Draw an energy level diagram showing the states involved in these transitions with state labels and transitions. Calculate the energy splitting between the ground state and these two low-lying excited states, and between the ground state and the $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^3 3\mathrm{s}^1$ state.
Hint: This part can be solved without any of the previous parts.

(d) When placed in a magnetic field, the ground state is Zeeman split in energy according to the m_j quantum number, obeying the Hamiltonian

$$H = -\vec{\mu}\cdot\vec{B} = g\mu_B B m_j = g\left(\frac{e\hbar}{2m_\mathrm{e}}\right)Bm_j, \qquad \text{where} \quad g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

How many levels does the ground state split into? Calculate the value of the the Landé g-factor for the ground state, and the energy splitting between neighboring levels in a magnetic field of $B = 1\,\mathrm{T}$.

(e) **Bonus:** Consider the Zeeman splitting according to the m_j quantum number of the other level besides the ground state from parts (b) and (c). Draw an energy level diagram showing the splitting of these four levels (no need to calculate the exact sizes of the splittings, but make sure the number of levels is correct). List and draw all allowed optical transitions $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^3 3\mathrm{s}^1 \rightarrow [\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^4$.
Hint: recall the selection rules for optical spectra: $\Delta S = 0$, $\Delta L = \pm 1$, $\Delta J = 0, \pm 1$, and $\Delta m_j = 0, \pm 1$, with the additional condition $m_j = 0 \nleftrightarrow 0$ if $\Delta J = 0$.

2. Decay chains (6 points)

Use only the values of Z (atomic number) and A (mass number) to calculate the number N_α (alpha) and N_{β^-} (beta) particles produced during the decay (i.e., equal to the number of α and β^- decays in the series, respectively) of the main naturally occurring ${}^{232}_{90}\mathrm{Th}$ (thorium-232) isotope to its stable end product ${}^{208}_{82}\mathrm{Pb}$ (lead-208).
Hint: Alpha decay is the emission of a ${}^4_2He$ nucleus, and β^- decay involves the process $n \rightarrow p + e^- + \overline{\nu}_e$. Write down how the atomic number Z and mass number A change in the two types of decay processes, and find an equation for N_α and N_{β^-}.

3. Stern-Gerlach experiment with a twist (2 + 4 + 3 + 2 = 11 points)
In the Stern-Gerlach experiment, a beam of neutral silver atoms ($S = J = 1/2$, $L = 0$) are passed through a magnetic field gradient in the $\hat{n} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ direction ($\hat{n}$ is a normalized vector pointing 45° from the x axis in the xy plane). Now the silver atoms are split according to their eigenvalues with respect to the operator

$$S_{\hat{n}} = \hat{n} \cdot \vec{S} = \frac{1}{\sqrt{2}}(S_x + S_y),$$

where

$$\hat{S}_x = \frac{\hbar}{2}\hat{\sigma}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{S}_y = \frac{\hbar}{2}\hat{\sigma}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2}\hat{\sigma}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(a) Show that the normalized eigenstate with $+\hbar/2$ eigenvalue is

$$|+\rangle = \frac{1}{2}\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix}.$$

(Check that it is an eigenvector, and that it is normalized.)

(b) Find the other normalized eigenstate $|-\rangle$ with $-\hbar/2$ eigenvalue.

(c) Calculate the expectation values of $\langle + | S_x | + \rangle$, $\langle + | S_y | + \rangle$, and $\langle + | S_z | + \rangle$.
Hint: Check that your result is consistent with the classical picture that the spin is polarized in the $\hat{n}$ direction.

(d) We direct the beam with $+\hat{n}$ polarization ($|+\rangle$ with $+\hbar/2$ eigenvalue) into another Stern-Gerlach setup, with the field gradient pointing in the z direction. What are the intensities of the two beams observed after the second setup relative to the intensity of the incoming beam?

4. Variational Methods: Harmonic oscillator (2 + 4 + 4 + 3 + 2 + 1 = 16 points)
Consider an electron in a one-dimensional harmonic potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2,$$

where $\omega = 8.88 \times 10^{25}\,\mathrm{s}^{-1}$. Consider a variational wavefunction, characterized by a single parameter α with nm units, and a normalization constant N:

$$\psi(r) = N\frac{1}{x^2 + \alpha^2}.$$

(a) Write the normalization condition and show that it is fulfilled for $N = \sqrt{\frac{2}{\pi}}\alpha^{3/2}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{1}{(x^2+\alpha^2)^2} = \frac{\pi}{2}\frac{1}{\alpha^3}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K \rangle_\psi$ that depends parametrically on α.
Hint: use the integral identities $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{1}{(x^2+\alpha^2)^3} = \frac{3}{8}\frac{\pi}{\alpha^5}$ *and* $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{x^2}{(x^2+\alpha^2)^4} = \frac{1}{16}\frac{\pi}{\alpha^5}$, *and don't forget the normalization constant.*

(c) Calculate the expression for the potential energy expectation value $\langle V \rangle_\psi$ that depends parametrically on α.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{x^2}{(x^2+\alpha^2)^2} = \frac{\pi}{2\alpha}$.

(d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H \rangle_\psi = \langle K \rangle_\psi + \langle V \rangle_\psi$, and apply the variational principle to find an equation for α in the ground state.

(e) Find the approximate solution α_0 and calculate its numerical value.

(f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$ and verify that $E_0 > \frac{1}{2}\hbar\omega$ (exact solution).

OFFICIAL SOLUTIONS

1. Energy spectrum of atomic Oxygen (5 + 4 + 5 + 3 + 6* = 17 + 6* points)
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$$H = -\vec{\mu}\cdot\vec{B} = g\mu_B B m_j = g\left(\frac{e\hbar}{2m_\mathrm{e}}\right)Bm_j, \qquad \text{where} \quad g = 1 + \frac{J(J+1)+S(S+1)-L(L+1)}{2J(J+1)}.$$

How many levels does the ground state split into? Calculate the value of the the Landé g-factor for the ground state, and the energy splitting between neighboring levels in a magnetic field of $B = 1\,\mathrm{T}$.

(e) **Bonus:** Consider the Zeeman splitting according to the m_j quantum number of the other level besides the ground state from parts (b) and (c). Draw an energy level diagram showing the splitting of these four levels (no need to calculate the exact sizes of the splittings, but make sure the number of levels is correct). List and draw all allowed optical transitions $[\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^3 3\mathrm{s}^1 \rightarrow [\mathrm{He}]2\mathrm{s}^2 2\mathrm{p}^4$.
Hint: recall the selection rules for optical spectra: $\Delta S = 0$, $\Delta L = \pm 1$, $\Delta J = 0, \pm 1$, *and* $\Delta m_j = 0, \pm 1$, *with the additional condition* $m_j = 0 \nrightarrow 0$ *if* $\Delta J = 0$.

Solution

(a) The p orbital has $\ell = 1$, so there are three available states corresponding to $m_\ell = +1, 0, -1$. To maximize S and L, the $m_\ell = +1$ state is doubly occupied by two electrons with opposite spin, and the other two states are occupied by a single spin up electron each. So the total quantum numbers are $L = 2\times 1 + 0 + (-1) = 1$ and $S = 3\times\frac{1}{2} - \frac{1}{2} = 1$. The shell is more than half-filled (there are doubly occupied orbitals), so J is maximized. In general $|L-S| \le J \le |L+S|$, so in the ground state $J = 2$. In the spectroscopic notation, the ground state is denoted as ${}^3P_2$.

(b) In the excited state, $L = 1 + 0 + (-1) + 0 = 0$ (the 3s orbital has $l = 0$), and $S = 3\times\frac{1}{2} - \frac{1}{2} = 1$. As $L = 0$, the only possible value is $J = 1$. In the spectroscopic notation, the state is denoted as ${}^3S_1$.

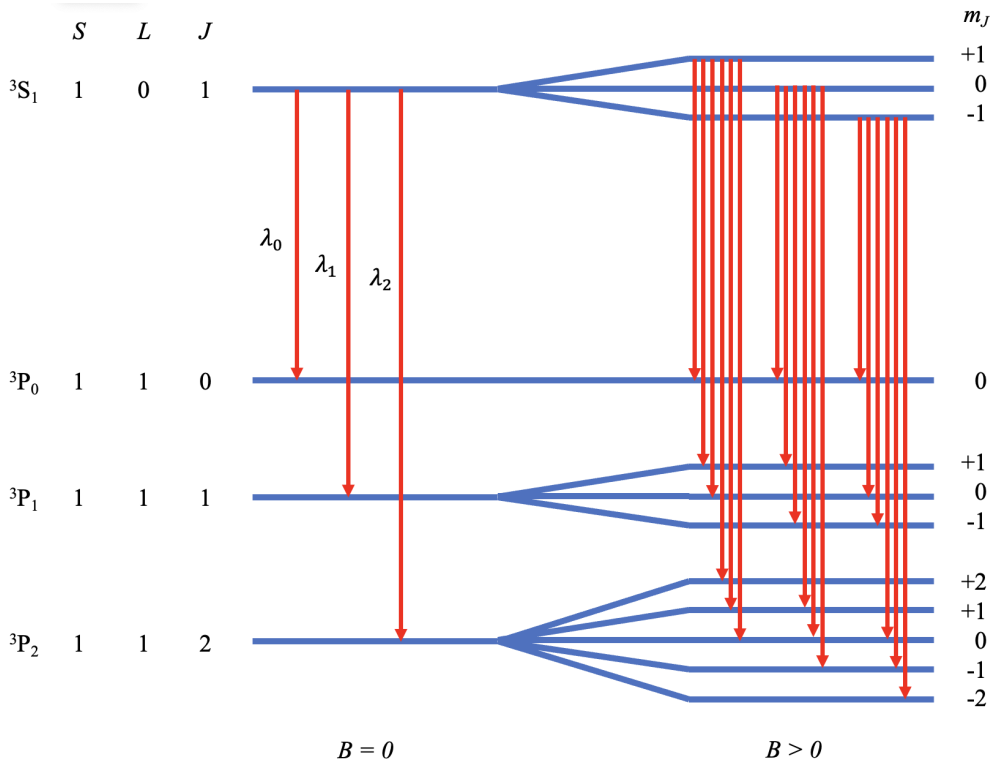


Figure 1: Energy level diagram of oxygen without and with a magnetic field.

(c) The level diagram without magnetic field and the transitions corresponding to the three spectral lines are shown in Fig. 1. The energy difference between two levels is given by the energy of the photon emitted during the transition

$$E = pc = \frac{hc}{\lambda}.$$

The energies of the levels compared to the ground state are given by

$$E_{^3S_1} = \frac{hc}{\lambda_2} = 9.521\,\text{eV}, \tag{1}$$

$$E_{^3P_1} = \frac{hc}{\lambda_2} - \frac{hc}{\lambda_1} = 19.70\,\text{meV}, \tag{2}$$

$$E_{^3P_0} = \frac{hc}{\lambda_2} - \frac{hc}{\lambda_0} = 27.70\,\text{meV}.$$

(d) The ground state has $J = 2$, and $-J \leq m_j \leq J$, so m_j can take the values $m_j = +2, +1, 0, -1, -2$. The levels shift in energy proportional to the m_j quantum number, so the ground state splits into 5 levels in magnetic field. Substituting $S = 1$, $L = 1$, and $J = 2$ for the ground state, we find $g = 3/2$. Adjacent levels have a difference in the m_j quantum number $\Delta m_j = 1$, so the energy difference is given by

$$\Delta E = g\mu_B B\,\Delta m_j = \frac{3}{2}\mu_B B = 86.83\,\mu\text{eV}.$$

(e) The energy diagram showing the anomalous Zeeman splitting and all allowed optical transitions is shown in Fig. 1. Every level is split according to its J quantum number into $2J + 1$ levels, and there are in total 18 transitions obeying the optical selection rules.

2. Decay chains (6 points)
Use only the values of Z (atomic number) and A (mass number) to calculate the number N_α (alpha) and N_{β^-} (beta) particles produced during the decay (i.e., equal to the number of α and β^- decays in the series, respectively) of the main naturally occurring ${}^{232}_{90}\mathrm{Th}$ (thorium-232) isotope to its stable end product ${}^{208}_{82}\mathrm{Pb}$ (lead-208).

Solution
The α decay is a nuclear disintegration process with the emission of an alpha particle

$$ {}^{A}_{Z}\mathrm{X} \rightarrow {}^{A-4}_{Z-2}\mathrm{Y} + {}^{4}_{2}\mathrm{He} $$

whereas the β^- decay consists of an emission of a high-energy electron from a nucleus

$$ {}^{A}_{Z}\mathrm{X} \rightarrow {}^{A}_{Z+1}\mathrm{Y} + {}^{0}_{-1}\mathrm{e}. $$

Therefore, by looking at the values of Z and A for the initial and stable product of the reaction, one has

$$ \begin{aligned} 232 - 4N_\alpha &= 208 \\ 90 - 2N_\alpha + N_{\beta^-} &= 82. \end{aligned} $$

The above equations admit only one solution, yielding that in the chain decay $N_\alpha = 6$ alpha particles and $N_{\beta^-} = 4$ beta particles are emitted.

For the sake of completeness, the full decay *Thorium series* is found in Fig. 2. Note that after decaying to Bismuth, there are two possible decay paths, the β^- path with 64.06 % probability and the α path with 35.94 % probability, which lead to the same stable product.

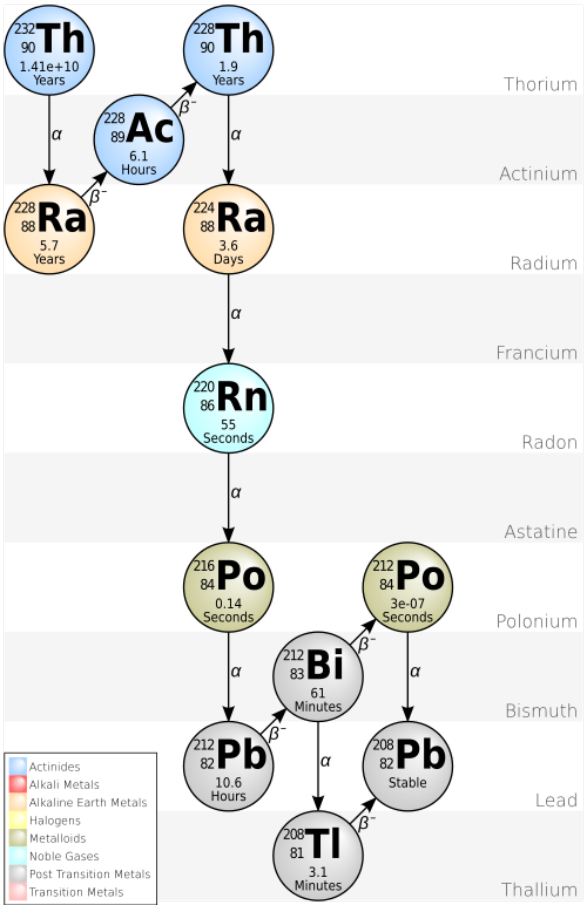


Figure 2: The $4n$ decay chain of ${}^{232}\mathrm{Th}$, commonly called the *"thorium series"*.
Source: `https://en.wikipedia.org/wiki/Thorium-232`.

3. Stern-Gerlach experiment with a twist (2 + 4 + 3 + 2 = 11 points)
In the Stern-Gerlach experiment, a beam of neutral silver atoms ($S = J = 1/2$, $L = 0$) are passed through a magnetic field gradient in the $\hat{n} = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right)$ direction ($\hat{n}$ is a normalized vector pointing 45° from the x axis in the xy plane). Now the silver atoms are split according to their eigenvalues with respect to the operator

$$S_{\hat{n}} = \hat{n} \cdot \vec{S} = \frac{1}{\sqrt{2}}(S_x + S_y),$$

where

$$\hat{S}_x = \frac{\hbar}{2}\hat{\sigma}_x = \frac{\hbar}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{S}_y = \frac{\hbar}{2}\hat{\sigma}_y = \frac{\hbar}{2}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{S}_z = \frac{\hbar}{2}\hat{\sigma}_z = \frac{\hbar}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(a) Show that the normalized eigenstate with $+\hbar/2$ eigenvalue is

$$|+\rangle = \frac{1}{2}\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix}.$$

(Check that it is an eigenvector, and that it is normalized.)

(b) Find the other normalized eigenstate $|-\rangle$ with $-\hbar/2$ eigenvalue.

(c) Calculate the expectation values of $\langle + | S_x | + \rangle$, $\langle + | S_y | + \rangle$, and $\langle + | S_z | + \rangle$.

(d) We direct the beam with $+\hat{n}$ polarization ($|+\rangle$ with $+\hbar/2$ eigenvalue) into another Stern-Gerlach setup, with the field gradient pointing in the z direction. What are the intensities of the two beams observed after the second setup relative to the intensity of the incoming beam?

Solution

(a) It can be easily verified that

$$\frac{1}{\sqrt{2}}(S_x + S_y)|+\rangle = \frac{\hbar}{4\sqrt{2}}\begin{pmatrix} 0 & 1-i \\ 1+i & 0 \end{pmatrix}\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{4\sqrt{2}}\begin{pmatrix} 2 \\ \sqrt{2}(1+i) \end{pmatrix} = \frac{\hbar}{2}|+\rangle.$$

and

$$\langle + | + \rangle = \frac{1}{4}\left(\sqrt{2},\ 1-i\right)\cdot\begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{1}{4}(2+2) = 1.$$

(b) We rescale the Hamiltonian by removing the $\hbar/2$ prefactor, and look for an eigenvector with -1 eigenvalue in the form

$$|-\rangle = \begin{pmatrix} u \\ v \end{pmatrix}.$$

The eigenvalue-eigenvector equation reads

$$\frac{1}{\sqrt{2}}(S_x + S_y)|-\rangle = -\frac{\hbar}{2}|-\rangle, \tag{3}$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1-i \\ 1+i & 0 \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix} = -\begin{pmatrix} u \\ v \end{pmatrix}, \tag{4}$$

$$\frac{1}{\sqrt{2}}\begin{pmatrix} (1-i)v \\ (1+i)u \end{pmatrix} = -\begin{pmatrix} u \\ v \end{pmatrix}.$$

From the second line we get $v = -\frac{1}{\sqrt{2}(1+i)}u$. Choosing $u = \sqrt{2}$, this results in the unnormalized eigenvector

$$|-\rangle = N \begin{pmatrix} \sqrt{2} \\ -(1+i) \end{pmatrix}.$$

The normalization condition reads

$$1 = \langle -|- \rangle = N^2 \begin{pmatrix} \sqrt{2}, & -(1-i) \end{pmatrix} \cdot \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = N^2(2+2) = 4N^2.$$

So $N = 1/2$, and the normalized eigenvector is

$$|-\rangle = \frac{1}{2} \begin{pmatrix} \sqrt{2} \\ -(1+i) \end{pmatrix}.$$

(c)

$$\langle + | S_x | + \rangle = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 1+i \\ \sqrt{2} \end{pmatrix} = \frac{\hbar}{8}(\sqrt{2}(1+i) + \sqrt{2}(1-i)) = \frac{\hbar}{2\sqrt{2}},$$

$$\langle + | S_y | + \rangle = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 1-i \\ i\sqrt{2} \end{pmatrix} = \frac{\hbar}{8}(\sqrt{2}(1-i) + \sqrt{2}(1+i)) = \frac{\hbar}{2\sqrt{2}},$$

$$\langle + | S_z | + \rangle = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ 1+i \end{pmatrix} = \frac{\hbar}{8} \begin{pmatrix} \sqrt{2}, & 1-i \end{pmatrix} \begin{pmatrix} \sqrt{2} \\ -(1+i) \end{pmatrix} = \frac{\hbar}{8}(2-2) = 0.$$

This result is consistent with the classical picture, that the spin vector has magnitude $\hbar/2$ and points in the direction of $\hat{n}$, and the spin expectation values are the same as the projections of this vector.

(d) The intensities of the two beams after the setup with z orientation are proportional to the probabilities of the spins pointing in the $\pm z$ direction. This probability is given by the squared magnitudes of components of the spin state given in the standard S_z basis (where the S_z operator is diagonal). Hence the probabilities in state $|+\rangle$ are:

$$P(+z) = \left| \frac{\sqrt{2}}{2} \right|^2 = \frac{1}{2},$$

$$P(-z) = \left| \frac{1+i}{2} \right|^2 = \frac{1}{2}.$$

So the relative intensities of the two beams are 1/2 and 1/2 of the incoming beam.

4. Variational Methods: Harmonic oscillator (2 + 4 + 4 + 3 + 2 + 1 = 16 points)
Consider an electron in a one-dimensional harmonic potential, described by the Hamiltonian

$$H = K + V = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + \frac{1}{2}m\omega^2 x^2,$$

where $\omega = 8.88 \times 10^{25}\,\mathrm{s}^{-1}$. Consider a variational wavefunction, characterized by a single parameter α with nm units, and a normalization constant N:

$$\psi(r) = N\frac{1}{x^2+\alpha^2}.$$

(a) Write the normalization condition and show that it is fulfilled for $N = \sqrt{\frac{2}{\pi}}\alpha^{3/2}$.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{1}{(x^2+\alpha^2)^2} = \frac{\pi}{2}\frac{1}{\alpha^3}$.

(b) Calculate the expression for the kinetic energy expectation value $\langle K\rangle_\psi$ that depends parametrically on α.
Hint: use the integral identities $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{1}{(x^2+\alpha^2)^3} = \frac{3}{8}\frac{\pi}{\alpha^5}$ *and* $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{x^2}{(x^2+\alpha^2)^4} = \frac{1}{16}\frac{\pi}{\alpha^5}$, *and don't forget the normalization constant.*

(c) Calculate the expression for the potential energy expectation value $\langle V\rangle_\psi$ that depends parametrically on α.
Hint: use the integral identity $\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{x^2}{(x^2+\alpha^2)^2} = \frac{\pi}{2\alpha}$.

(d) Calculate the expression for the energy expectation value $E(\alpha) = \langle H\rangle_\psi = \langle K\rangle_\psi + \langle V\rangle_\psi$, and apply the variational principle to find an equation for α in the ground state.

(e) Find the approximate solution α_0 and calculate its numerical value.

(f) Calculate the approximate ground state energy $E_0 = E(\alpha_0)$ and verify that $E_0 > \frac{1}{2}\hbar\omega$ (exact solution).

Solution

(a) The normalization condition reads

$$1 = \int_{-\infty}^{\infty} \mathrm{d}x\, \psi^*(x)\psi(x) = N^2 \int_{-\infty}^{\infty} \mathrm{d}x\, \frac{1}{(x^2+\alpha^2)^2} = N^2\frac{\pi}{2}\alpha^{-3}$$

thus yielding

$$N = \sqrt{\frac{2}{\pi}}\alpha^{3/2}.$$

(b) The expectation value of the kinetic energy is given by

$$\langle\psi|\, K\,|\psi\rangle = -\frac{\hbar^2}{2m}\frac{2}{\pi}\alpha^3 \int_{-\infty}^{\infty} \mathrm{d}x\, \frac{1}{x^2+\alpha^2}\left(\frac{\partial^2}{\partial x^2}\frac{1}{x^2+\alpha^2}\right) = -\frac{\hbar^2}{2m}\frac{2}{\pi}\alpha^3 \int_{-\infty}^{\infty} \mathrm{d}x\, \left[-\frac{2}{(x^2+\alpha^2)^3} + \frac{8x^2}{(x^2+\alpha^2)^4}\right] = \frac{\hbar^2}{4m}\frac{1}{\alpha^2}$$

(c) The expectation value of the potential energy is given by

$$\langle\psi|\, V\,|\psi\rangle = \frac{1}{2}m\omega^2\frac{2}{\pi}\alpha^3 \int_{-\infty}^{\infty} \mathrm{d}x\, \frac{x^2}{(x^2+\alpha^2)^2} = \frac{1}{2}m\omega^2\alpha^2$$

(d) The parametric expression for ground state energy reads

$$E(\alpha) = \frac{\hbar^2}{4m}\frac{1}{\alpha^2} + \frac{1}{2}m\omega^2\alpha^2,$$

for which one can verify that each term has the proper physical dimensions. The variational principle implies that

$$\frac{d}{d\alpha}E(\alpha) = -\frac{\hbar^2}{2m}\frac{1}{\alpha^3} + m\omega^2\alpha = 0$$

which admits the solution

$$\alpha_0 = \left(\frac{\hbar^2}{2m^2\omega^2}\right)^{1/4} = 9.60 \times 10^{-10}\,\mathrm{m}.$$

(e) Substituting the optimal value $\alpha = \alpha_0$ in the parametric expression for the ground state energy, one finds

$$E_0 = E(\alpha_0) = \frac{\sqrt{2}}{2}\hbar\omega = 0.260\,\mathrm{eV} > \frac{1}{2}\hbar\omega = 0.184\,\mathrm{eV}.$$